Acid-Base Chemistry

Acid - Base Definitions

- Brønsted-Lowry Acid/Base Definition
  - Acid: compound that donates a H^+ to another species
  - Base: compound that accepts a H^+ from another compound

- Strong Acid/Base - completely transfers its protons to/from water (completely dissociated)
- Weak Acid/Base - partially transfers its protons to water (some compounds remain unchanged)

Strong Acids and Bases

Strong Acids (completely dissociate)	
HCl	Hydrochloric acid
HBr	Hydrobromic acid
HI	Hydroiodic acid
$HClO_4$	Perchloric acid
HNO_3	Nitric acid
H_2SO_4	Sulfuric acid

Strong Bases (completely dissociate)	
LiOH	Lithium hydroxide
NaOH	Sodium hydroxide
KOH	Potassium hydroxide
$Ca(OH)_2$	Calcium hydroxide
$Sr(OH)_2$	Strontium hydroxide
$Ba(OH)_2$	Barium hydroxide

$pH = -\log[H^+]$ or
$pH = -\log[H_3O^+]$

$[H_3O^+] = 10^{-pH}$

Ex: $HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$

Note: irreversible arrow!

$NaOH(aq) \rightarrow Na^+(aq) + OH^-(aq)$

$pOH = -\log[OH^-]$

$[OH^-] = 10^{-pOH}$

Weak Acids

Incomplete Transfer of Proton in Solution
(compound does not completely ionize)

$HA(aq) + H_2O(l) \rightleftharpoons A^-(aq) + H_3O^+(aq)$

$$K_{eq} = \frac{[A^-][H_3O^+]}{[HA]}$$

$pK_a = -\log K_a$

Can be used as a measure of acid strength

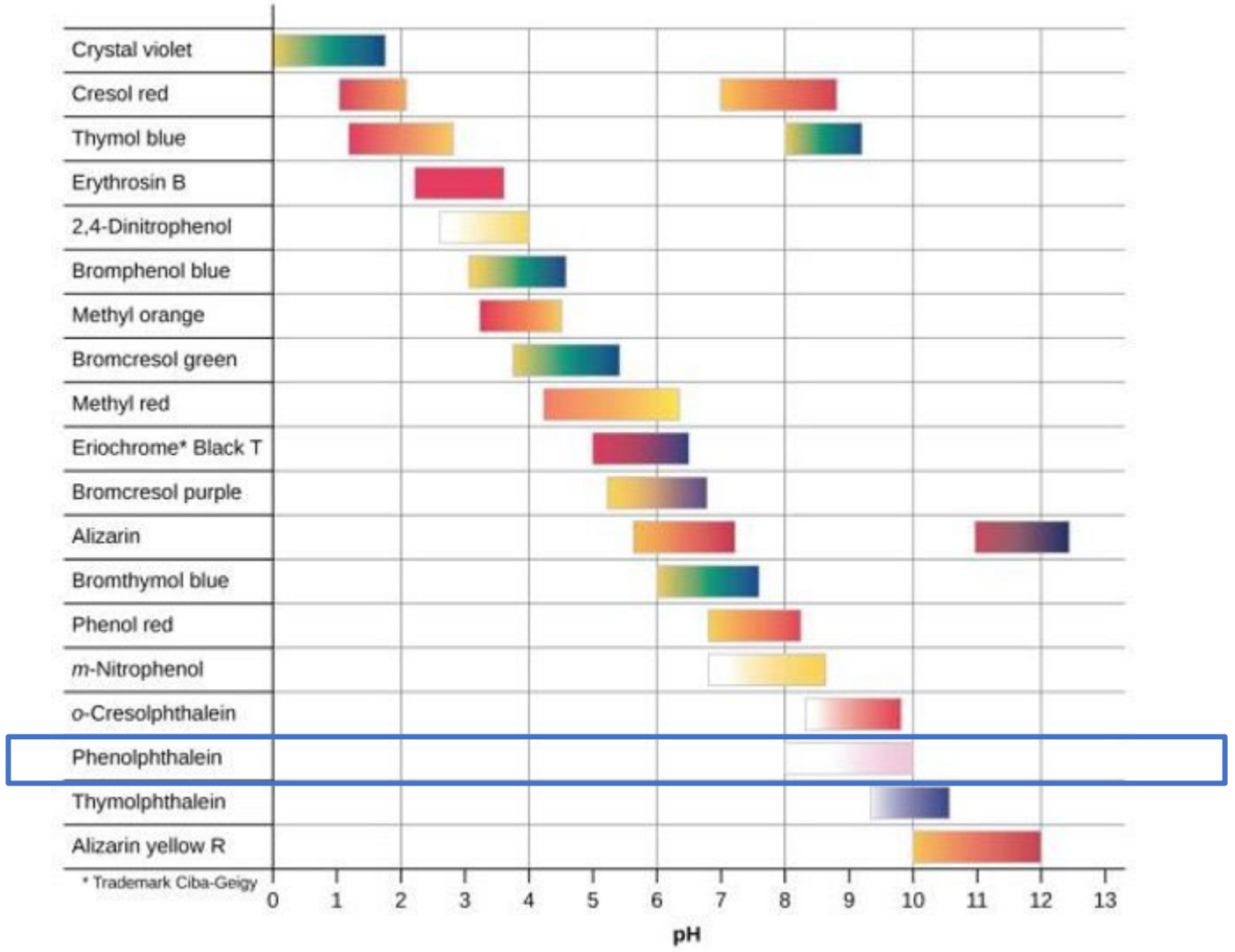
$$K_a = \frac{[A^-][H_3O^+]}{[HA]}$$

(acid dissociation constant)

Titrations

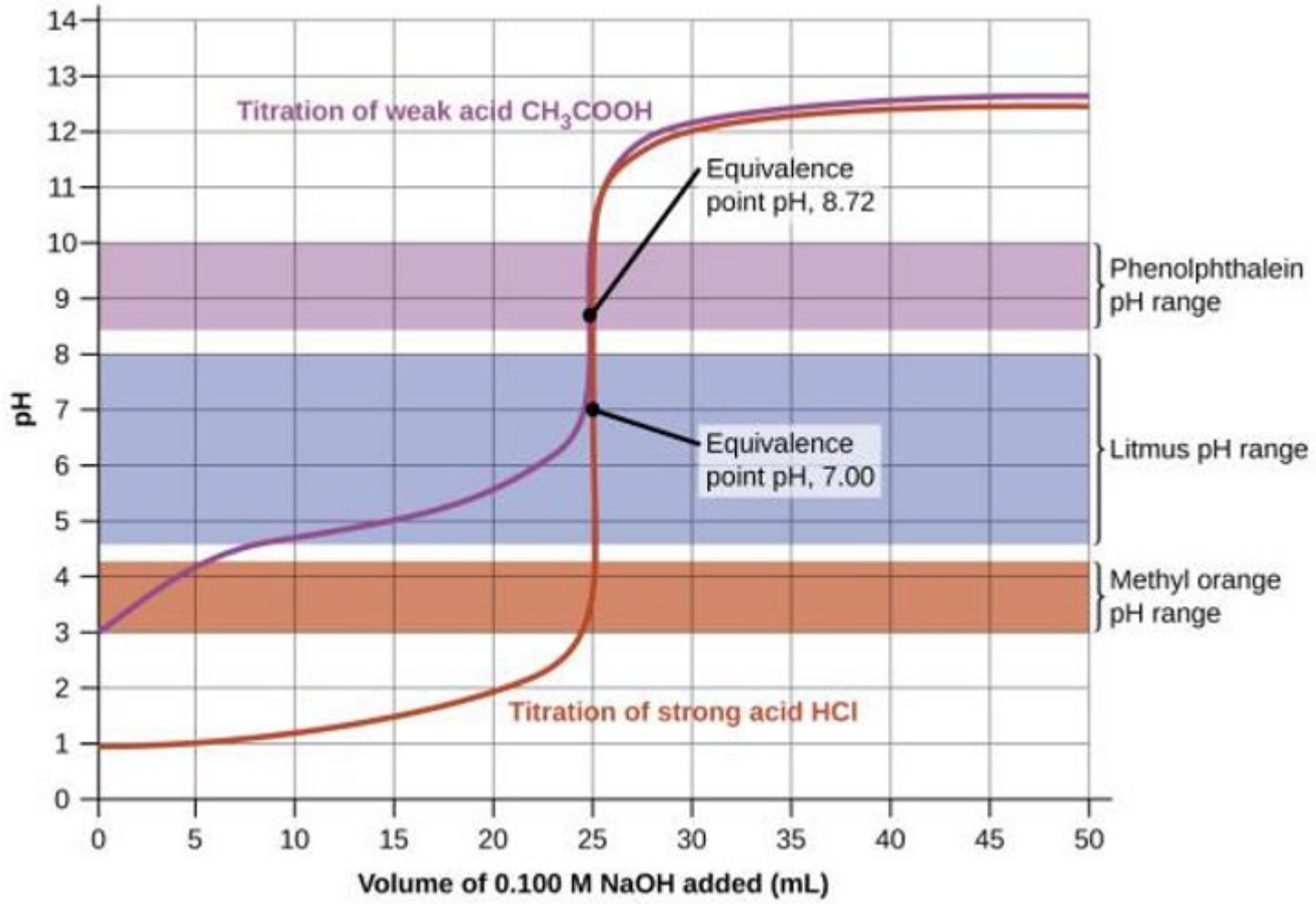
- titration - addition of a known concentration of a base to an unknown concentration of an acid (or vice versa)
- equivalence point - point at which stoichiometrically equivalent quantities of acid and base have been brought together
- indicator - a compound that changes color at a specific pH (used to help indicate the equivalence point)

Indicators



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Titration Curves – Weak Acid + Strong Base



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$$2\,H_2O(aq) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$$

$$HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

This is the key to this lab!!

When $[A^-] = [HA] \rightarrow \frac{1}{2}$ *Neutralized*

$$[H^+] = K_a \Longrightarrow pH = pK_a$$

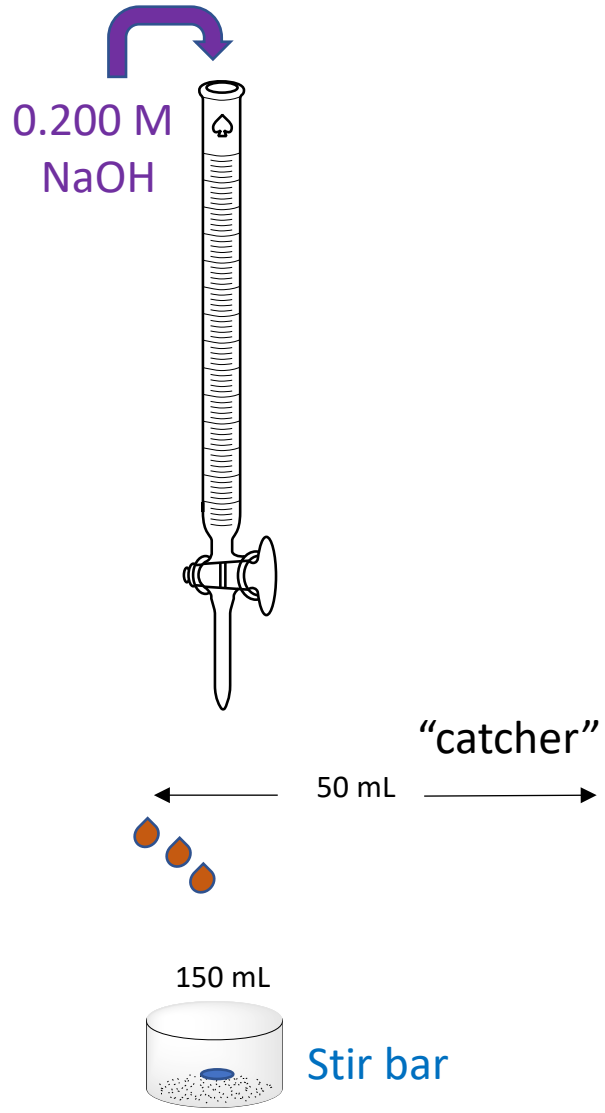
$$pK_a = -\log(K_a)$$

Procedure for Part A:

4 ➞ 7 Calibration → pH 4, pH 10

8. Get Unknown (from cart)

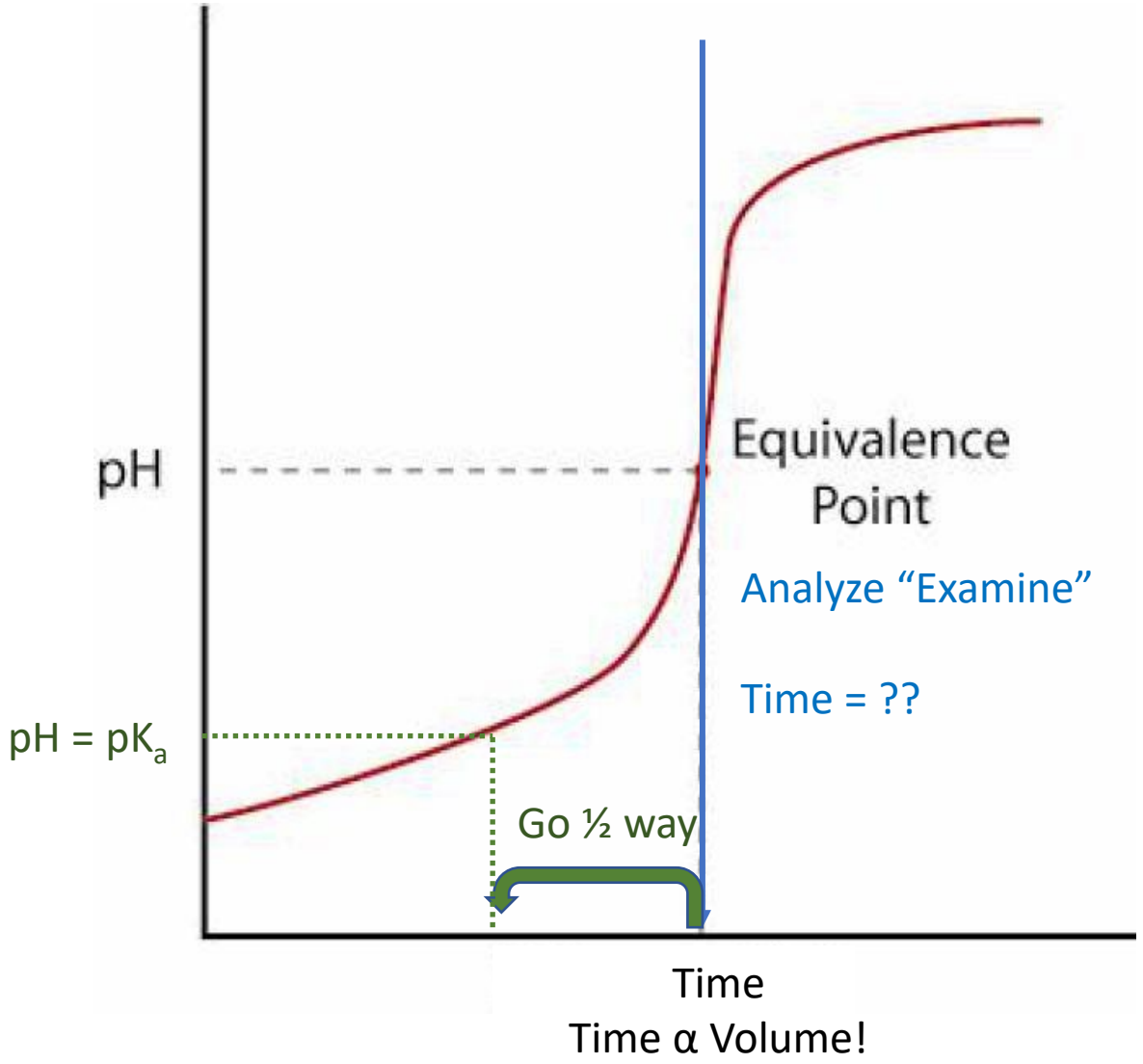
Make sure to record the unknown #



Into 150 mL Beaker:

1. 20 mL H_2O
2. 20 mL Unknown Weak Acid
3. 3 drops indicator

Data Collection for Part A:



Do this Twice!!

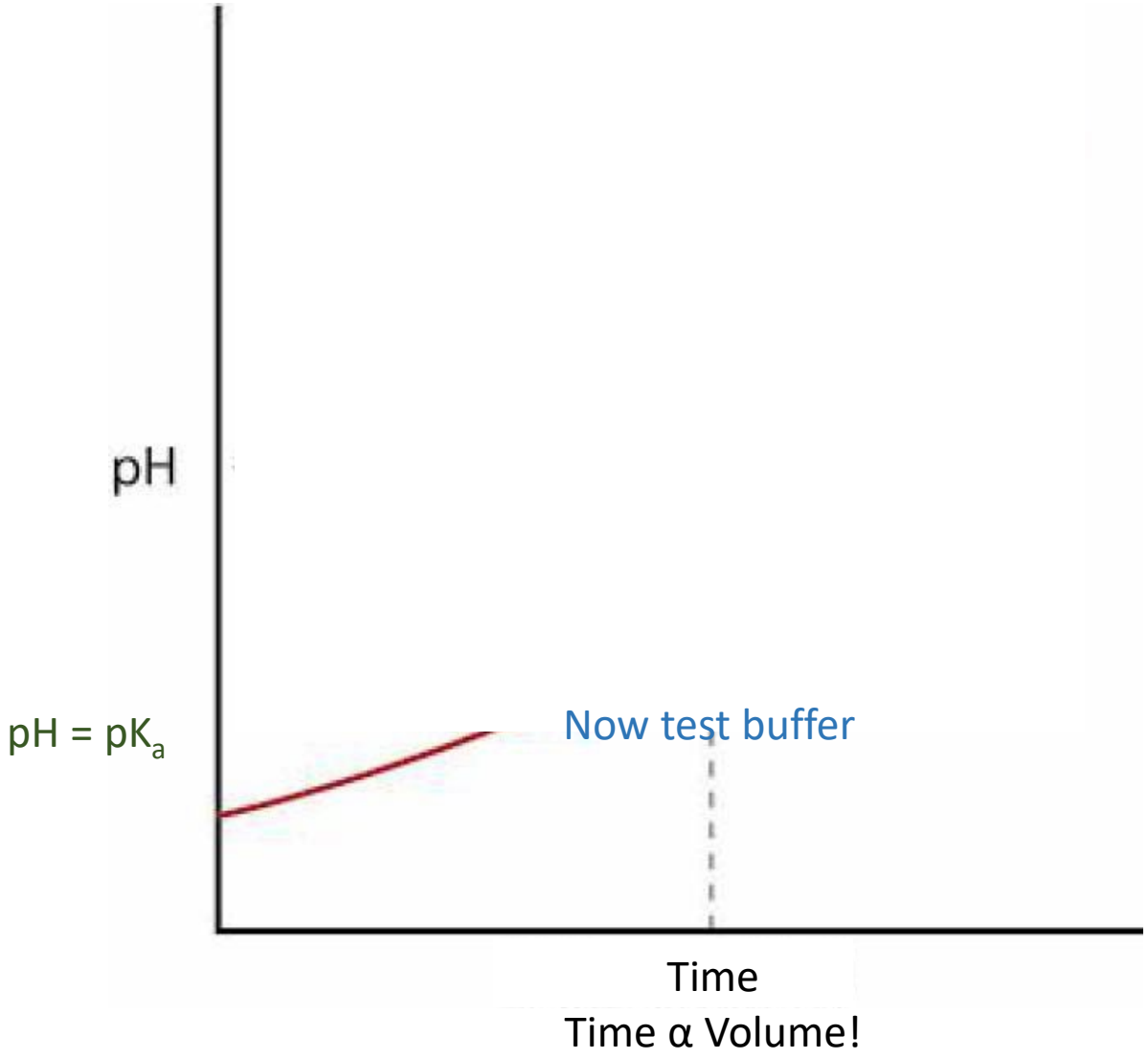
Use the Table in your manual to identify your unknown acid based on the K_a values.

Buffers

Buffers work best when you have a 'large' amount of HA and A^- present. The amount of acid or base a buffer can neutralize is called the **buffer capacity**.

Because you want equal amounts of HA and A^- present – the best pH for buffers to work is equal to their K_a

Procedure for Part B:



Repeat Procedure A, but
STOP adding NaOH
when pH = pK_a

Test the buffer capacity of water

Procedure for Part C:

0.1 M

MW/FW

1. $100\ mL\left(\frac{1L}{1000\ mL}\right)\left(\frac{0.1\ mole}{1L}\right)\left(\frac{___g\ \text{“}salt\text{”}}{1\ mole\ \text{“}salt\text{”}}\right) =$ g salt to use
2. ➞ 6.
7. Reaction responsible:
   - Let's say your salt was $MgBr_2$
   - $MgBr_2(s) \xrightarrow{H_2O} Mg^{2+}(aq) + 2\ Br^-(aq)$
   - $Mg^{2+}(aq) + 2\ H_2O(l) \leftrightarrow Mg(OH)_2(aq) + 2H^+(aq)$

     Conj. acid — Weak base — Strong acid

     ➞ ACIDIC!
   - $Br^-(aq) + H_2O(l) \leftrightarrow HBr(aq) + OH^-(aq)$

     Conj. base — Strong acid — Strong base

     ➞ NEUTRAL!

OVERALL: ACIDIC!!

Safety and Clean-up

- Be careful with the acids and bases – these will cause chemical burns
  - If any spills occur, please let us know so we can get them cleaned up properly
- Dispose of all waste into the appropriate waste containers
- Please return the remaining amount of your unknown weak acid from part A to the cart (others will be using these)
- Be sure to upload your lab notebook!